Bernoulli Types

An Algebra of Approximate Computation

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April 29, 2026

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The Python companion library `bernoulli` is released separately under the MIT License.

ORCID: `0000-0001-6443-9897`
Web: `https://metafunctor.com`
Source: `https://github.com/queelius/bernoulli-book`

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Preface

This is a placeholder preface. The final preface will:

- State the book's thesis and audience.
- Declare prerequisites: undergraduate probability, basic discrete math, comfort with proof, working Python knowledge for the notebook companion.
- Explain the lockstep notebook discipline (some chapters carry notebooks; theory-heavy chapters do not, by design).
- Provide a reading guide for the smart-undergrad-upward reader (chapters that admit a soft landing) and the adjacent-field grad student (the modal reader).

Part I

The Atom

Chapter 1

The Noisy Bit

Placeholder. Content drafted in Plan 1. The framework appears in [Tow26].

Chapter 2

Binary Channels and Model Order

Placeholder. Content drafted in Plan 1.

Chapter 3

Boolean Algebra Under Noise

Placeholder. Content drafted in Plan 1.

Chapter 4

Two Axioms and Kronecker Factorization

Placeholder. Content drafted in Plan 1.

Part II

From Bits to Sets

Chapter 5

Approximate Sets and Predicates

Placeholder. Content drafted in Plan 2.

Chapter 6

Composition of Sets

Placeholder. Content drafted in Plan 2.

Chapter 7

Classification Measures

Placeholder. Content drafted in Plan 2.

Chapter 8

Information-Theoretic Bounds

Placeholder. Content drafted in Plan 2.

Part III

From Sets to Maps

Chapter 9

Approximate Functions and n-ary Channels

Placeholder. Content drafted in Plan 3.

Chapter 10

Sketches and Estimators

Placeholder. Content drafted in Plan 3.

Chapter 11

Randomized Algorithms

Placeholder. Content drafted in Plan 3.

Chapter 12

Approximate Relations

Placeholder. Content drafted in Plan 3.

Chapter 13

The Algebra of Approximate Types

Placeholder. Content drafted in Plan 3.

Part IV

Optimality and Construction

Chapter 14

Optimal Bernoulli Sets

Placeholder. Content drafted in Plan 4.

Chapter 15

Perfect Hash Filters

Placeholder. Content drafted in Plan 4.

Part V

The Boundary and Beyond

Chapter 16

Approximate but Not Bernoulli

Placeholder. Content drafted in Plan 4.

Chapter 17

Cipher Maps and Trapdoor Computing

Placeholder. Content drafted in Plan 4.

Chapter 18

Encrypted Search as Relaxed Trapdoor

Placeholder. Content drafted in Plan 4.

Appendix A

Probability Refresher

Placeholder. Content drafted in Plan 5.

Appendix B

The Five Core Principles

Placeholder. Cheat-sheet of the Two Axioms, Parametric Parsimony, Space-Accuracy Duality, Composition Closure, and the Boolean Thread. Content drafted in Plan 5.

Appendix C

Library Walkthrough

Placeholder. Tour of the `bernoulli` *Python package as a unified entry point. Content drafted in Plan 5.*

Appendix D

Selected Exercise Solutions

Placeholder. Roughly half of the end-of-chapter exercises are worked here; the rest are reader practice. Content drafted in Plan 5.